Course Code	DESIGN THINKING & INNOVATION	L	T	P	Credits
		0	1	3	2.5

Course Objectives

- Introduce students to the principles, mindset and methodologies of **Design Thinking and human-centered problem solving**.
- Develop students' ability to **observe, empathize, analyze user needs and formulate well-defined engineering and societal problems**.
- Develop students' **creative and innovative thinking abilities** through systematic ideation, concept generation and evaluation techniques.
- Enable students to **develop, prototype, test and iteratively improve solutions** using user feedback and appropriate physical/digital tools.
- Develop students' ability to **identify innovation opportunities, create value propositions, understand basic IPR and communicate innovative solutions effectively**.

CO1	Explain the principles, processes and mindset of Design Thinking for engineering problem-solving and innovation	K2
CO2	Apply empathy, observation and problem-framing techniques to identify user needs and define engineering problems	K3
CO3	Apply creative and systematic ideation techniques to generate and select innovative solution concepts	K3
CO4	Analyze solution concepts and prototypes using user feedback and appropriate design criteria for iterative improvement	K4
CO5	Analyze the innovation potential of solutions considering value proposition, feasibility, sustainability, societal impact and basic IPR aspects	K4

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	1	–	–	1	–	–	1	1	2			
CO2	2	3	3	1	1	3	2	1	2	2	2			
CO3	1	2	3	1	2	1	1	–	3	2	2			
CO4	3	3	3	2	3	2	2	1	3	2	2			
CO5	1	2	3	1	2	3	3	2	2	3	3			

UNIT	COURSE CONTENT	Contact Hours
UNIT-I	**Design Thinking Foundations & Empathy** Introduction to Design Thinking ; Design thinking mindset and principles ; Design thinking vs. conventional engineering problem-solving ; Human-centered design ; Five-stage design thinking process ; Creativity, curiosity and observation ; Understanding users and stakeholders ; Empathy and user research ; Interviews, observation and contextual inquiry ; Empathy mapping ; Personas and user journeys ; Identifying explicit and latent/unarticulated needs ; Ethical considerations in user research **Practical Activities** • Observation of a real-life problem • Conducting user interviews • Preparing an empathy map • Developing a persona • Preparing a user journey map **Unit Outcome** Students will be able to **understand users, identify their needs and translate observations into meaningful design opportunities**.	
UNIT-II	**Problem Definition, Analysis & Opportunity Identification** From observations to insights ; Data synthesis and affinity mapping ; Identifying patterns and pain points ; Problem statements ; Point-of-View (POV) statements ; "How Might We" (HMW) questions ; Stakeholder mapping ; 5 Whys and root-cause analysis ; Problem prioritization ; Need–Opportunity mapping ; Problem validation ; Design constraints and specifications ; Social, environmental and sustainability considerations ; Identifying opportunities for innovation **Practical Activities** • Affinity mapping of interview data • Develop three alternative problem statements • Prepare POV and HMW statements • Stakeholder map • Root-cause analysis • Select and validate one problem/opportunity **Unit Outcome**	

	Students will be able to **analyze user research, identify root causes, frame well-defined problems and recognize opportunities for innovative solutions**.	
UNIT-III	**Creativity, Ideation & Concept Development** Creativity and innovation mindset; Divergent and convergent thinking ; Brainstorming and brainwriting ; SCAMPER ; Mind mapping ; Reverse brainstorming ; Analogical thinking ; Biomimicry ; Lateral thinking ; Forced connections ; Morphological analysis ; TRIZ — basic principles ; Individual and collaborative ideation ; AI-assisted ideation: opportunities and limitations ; Idea clustering and screening ; Impact–feasibility matrix ; Concept selection ; Design criteria and decision matrix **Practical Activities** • Generate 20–30 ideas for a selected problem • SCAMPER exercise • Brainwriting exercise • Biomimicry-based ideation • AI-assisted idea generation and critical evaluation • Prepare an idea-selection matrix • Select the top 2–3 concepts **Unit Outcome** Students will be able to **generate diverse solution concepts using creative techniques and systematically select promising ideas based on defined criteria**.	
UNIT-IV	**Prototyping, Experimentation & Validation** Concept development ; Prototype mindset ; Types of prototypes: Paper prototype, Physical prototype, Digital prototype, Storyboard, Role-play, Service prototype ; Low-fidelity and high-fidelity prototypes ; Rapid prototyping ; "Fail fast, learn fast" ; Minimum Viable Prototype (MVP) ; Design experimentation ; Prototype evaluation ; User testing ; Think-aloud method ; Feedback collection ; Usability and functionality testing; Iterative design ; Design critique ; Documentation of iterations ; Use of CAD, simulation and AI-assisted visualization tools **Practical Activities** • Develop a low-fidelity prototype • Prepare a storyboard • Build a physical/digital prototype • Conduct user testing • Collect structured feedback • Modify the prototype • Prepare an iteration log	
UNIT-V	**Innovation, Value Creation & Communication** **Innovation: meaning, characteristics and importance:** Invention vs. innovation ; Incremental, disruptive and radical innovation ; Product, process, service and social innovation ; Technology-driven vs. need-	

	driven innovation ; Sustainable and frugal innovation ; Innovation opportunities and emerging technologies ; From prototype to innovation ; Desirability, feasibility and viability ; Value Proposition ; Value Proposition Canvas ; Introduction to Business Model Canvas ; Customer segments and stakeholder value; Innovation validation; Market/user validation ; Intellectual Property Rights — introduction; Patents, copyright, trademarks and design protection ; Responsible innovation ; Inclusive and sustainable innovation ; Innovation ecosystem and entrepreneurship ; Communicating innovative ideas ; Innovation pitch **Practical Activities** 1. Classify existing products/services as incremental, disruptive or radical innovations. 2. Identify three innovation opportunities from a campus/community problem. 3. Prepare a **Value Proposition Canvas**. 4. Validate the proposed innovation with potential users. 5. Prepare an introductory **Business Model Canvas**. 6. Identify possible IPR protection. 7. Prepare and deliver a **3-minute Innovation Pitch**.	

Text Books

1. Chandan Adhikari, Prabir Kumar Das & Tanay Pramanik, Textbook of Design Thinking and Innovation, S. Chand Publishing, 2026.
2. Shalini Rahul Tiwari & Rohit Rajendra Swarup, Design Thinking: A Comprehensive Textbook, Wiley India, 2024

Reference Books

1. Soni, Pavan, Design Your Thinking: The Mindsets, Toolsets and Skill Sets for Creative Problem-Solving, Penguin Random House India.
2. Jauhari, Vinnie & Sudhanshu Bhushan, Innovation Management, Oxford University Press India.

Web References

1. Atal Innovation Mission, NITI Aayog – Innovation and Design Thinking Resources. https://aim.gov.in/
2. Institution's Innovation Council (IIC), Ministry of Education, Government of India, https://iic.mic.gov.in/
3. Startup India, Government of India – Innovation and Entrepreneurship Ecosystem, https://www.startupindia.gov.in/
4. National Innovation Foundation – India – Grassroots and Frugal Innovation, https://nif.org.in/
5. Stanford d.school – Design Thinking Bootleg and Design Thinking Resources, https://dschool.stanford.edu/resources/design-thinking-bootleg

6. IDEO.org – Human-Centered Design and Social Innovation Resources, https://www.ideo.org/
7. IDEO U – Design Thinking and Innovation Resources, https://www.ideou.com/pages/design-thinking
8. NPTEL – Online Courses on Design, Innovation, Creativity and Entrepreneurship, https://nptel.ac.in/
9. SWAYAM – Online Courses on Innovation, Entrepreneurship and Design Thinking, https://swayam.gov.in/